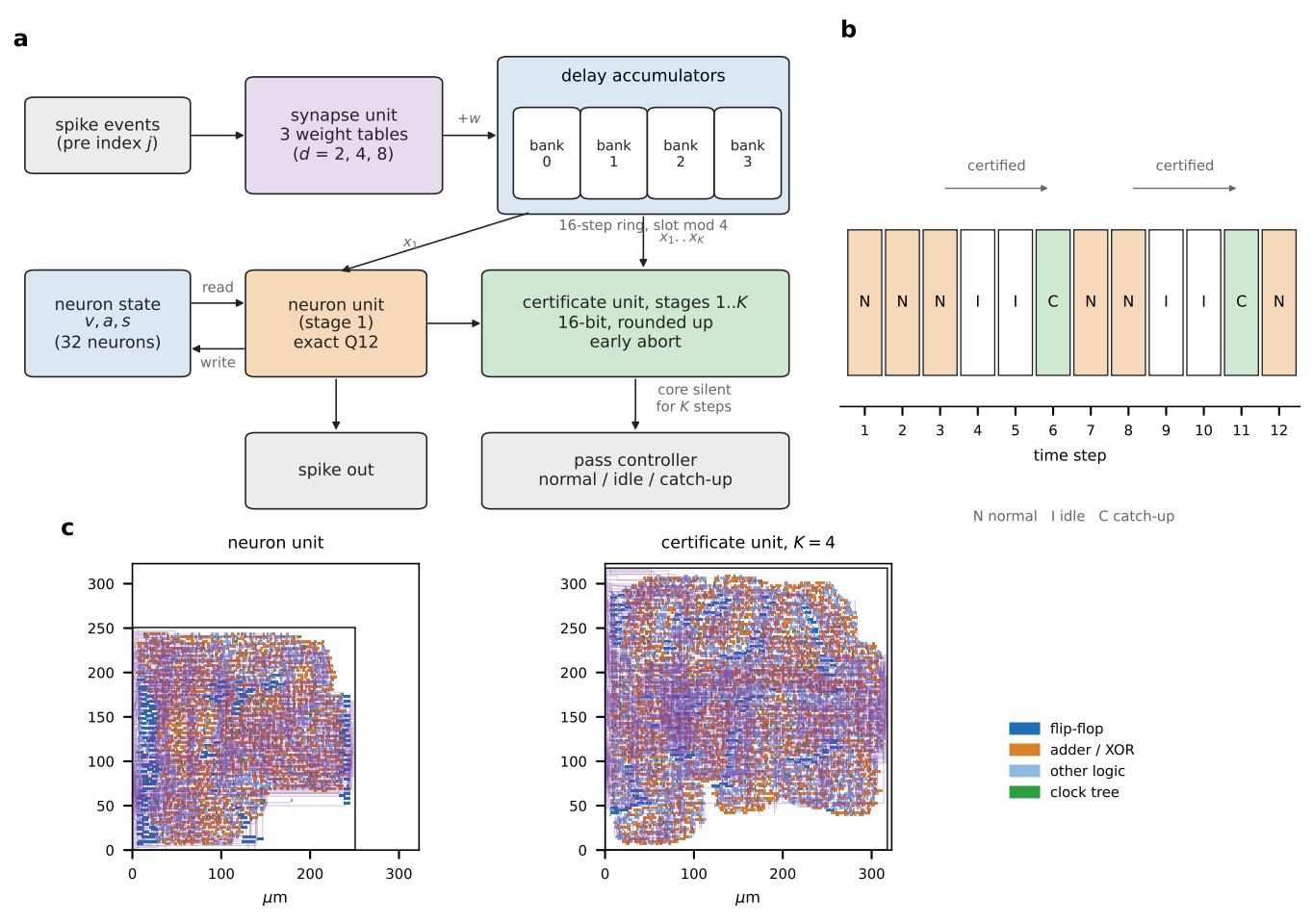

a
spike events
(pre index j)
synapse unit
3 weight tables
(d = 2, 4, 8)
+w
delay accumulators
bank 0
bank 1
bank 2
bank 3
16-step ring, slot mod 4
x1
x1..xK
neuron state
v, a, s
(32 neurons)
read
write
neuron unit
(stage 1)
exact Q12
certificate unit, stages 1..K
16-bit, rounded up
early abort
core silent
for K steps
spike out
pass controller
normal / idle / catch-up
b
certified
certified
N N N I I C N N I I C N
1 2 3 4 5 6 7 8 9 10 11 12
time step
N normal I idle C catch-up
c
neuron unit
certificate unit, K = 4
300
250
200
150
100
50
0
0 100 200 300
μm
flip-flop
adder / XOR
other logic
clock tree